



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Singly Linked List-Nodes

SINGLY LINKED LIST-NODES

"A hospital maintains patient records using a linked list. To generate reports, the system needs to determine the total number of records currently stored. Write a C program to count and display the total number of nodes in a singly linked list.

Test Case 1

Input

10 → 20 → 30 → 40 → 50

Output

Total number of nodes = 5"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 struct Node* createNode(int val) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11     newNode->data = val;
12     newNode->next = NULL;
13     return newNode;
14 }
15
16 void append(struct Node **head, int val) {
17     struct Node *newNode = createNode(val);
18
19     if (*head == NULL) {
20         *head = newNode;
21     }
22 }
```

OUTPUT

Total number of nodes = 10

INPUT

10 20 30 40 50

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